

BEMELINGUE Wilfried

Île-de-France, France

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AI & Data Engineering Student – Looking for a 6-month end-of-studies internship

PROFILE

Final-year engineering student (M2) at EPITA, with a strong analytical background and a deep interest in artificial intelligence and data. Known for adaptability, intellectual curiosity, and rigor in problem-solving. Experience on complex projects in collaborative environments, with the ability to clearly communicate technical concepts. Available for a 6-month internship starting immediately.

PROFESSIONAL EXPERIENCE

Intern – Data Pipeline Optimization, COMUTITRES (Paris) 09/2024 – 02/2025

- Migration and optimization of Python/SQL workflows to Airflow on AWS to improve processing reliability.
- Design of automatic retry and monitoring mechanisms using JSON configurations.
- Creation of KPI dashboards for Business Intelligence and improvement of CI/CD pipelines with GitLab.

Software Developer Intern, TDA CERMAI (Paris) 05/2022 – 07/2022 & 04/2023 – 07/2023

- Development of full-stack monitoring applications (JavaScript front-end, Python/ObjectScript back-end).
- Integration of REST APIs for real-time processing and display of operational data.
- Contribution to improving system stability and application performance.

ACADEMIC AND PERSONAL PROJECTS

Neuro-Graph Trader – Autonomous GNN-RL Agent — Extension of a Differential Graph Transformer architecture (Stanford CS224W) into a multi-agent decision-making system combining dynamic GNNs, reinforcement learning (PPO) and NLP on time-series data, with spatio-temporal attention, XAI modules and a complete modular training and evaluation pipeline. (*Python, PyTorch, PyG, FinRL*)

LLM Society – Multi-Agent LLM Framework — Simulation of complex interaction dynamics with local LLMs (Ollama: phi, gemma3): each agent has a configurable state (YAML) and interacts through structured JSON actions. Evaluation of coherence and emergent behaviors across multiple scenarios. (*Python, Ollama, PyYAML, multi-agents*)

Real-Time Anomaly Detection with GNN — Detection of anomalous patterns in large-scale transaction networks using real-time data ingestion (Spark), node classification via GraphSAGE/GCN and an adversarial RL approach (detection agent vs evasion agent). Explainability via GNNExplainer with subgraph generation for human analysts. (*Python, PyTorch Geometric, Spark*)

NLP-Based Document Filter — Automated relevance scoring of text documents through keyword extraction, TF-IDF vectorization and ML classification, enabling efficient matching between documents and structured queries. (*Python, NLTK, scikit-learn, pandas*)

EDUCATION

EPITA – Engineering Student (M2), SCIA-G specialization (AI and Graphs) 2023 – 2026

- Advanced courses in artificial intelligence, machine learning, distributed systems and data engineering.
- Applied projects in graph learning, optimization and scalable systems.
- Intensive training through programming, mathematics, cloud and engineering fundamentals bootcamps.

ECE – Paris (B2) 2022 – 2023

- Web development (PHP, JavaScript), cloud computing (AWS) and Java application development.

EPSI – Rennes (B1) 2021 – 2022

- Web application development (Python, PHP), SEO and Linux/Windows server administration.

Université Saint Jean – Yaoundé – Preparatory class 2020 – 2021

TECHNICAL SKILLS

Machine Learning & AI : Deep Learning, GNN, Reinforcement Learning, XAI, Reco. Systems, LLM, NLP

Data Engineering : Python, SQL, pandas, scikit-learn, Airflow, Spark, Data Pipelines

Cloud & DevOps : AWS, Docker, Kubernetes, GitLab CI/CD

Software Development : Python, Java, JavaScript/TypeScript, C/C++, REST API, React, FastAPI, Flask

Databases : MySQL, PostgreSQL, MongoDB, Neo4j

Visualization : Tableau, Power BI, Matplotlib, Seaborn

Mathematical Foundations : Linear Algebra, Probability Theory, Optimization, Stochastic Processes

Tools : Linux, Git, Jupyter, Bash, Hugging Face

LANGUAGES

French : Native language

English : Fluent (B2)